MDDS and Decoding on Center-out reach Task

Diletta Bartolini[1], Icaro Lorran Lopes Costa[2], Matteo Pizzinga[3], Mehdi Borjkhani[4]

Abstract

Understanding how neural population activity in motor cortex encodes movement parameters is fundamental to developing effective brain-computer interfaces (BCIs). We analyzed neural recordings from primary motor cortex (M1) of a monkey performing a center-out reach task to eight target directions using a joystick. Neural activity was examined using dimensionality reduction techniques including Principal Component Analysis (PCA), Uniform Manifold Approximation and Projection (UMAP), and Manifold Discovery via Dynamical Systems (MDDS) to uncover low-dimensional structure. We evaluated the effectiveness of these representations for target classification using Linear Discriminant Analysis and kinematic decoding via linear regression. MDDS revealed a cone-like neural manifold structure and significantly outperformed other methods, achieving Matthews correlation coefficients of $\rho_M \approx 0.87$ for target classification compared to $\rho_M \approx 0.47$ for raw neural data. Similarly, MDDS-generated representations improved kinematic decoding with linear regression performance ($R^2 \approx 0.66$ vs $R^2 \approx 0.57$ for raw data). A simple Brain-Computer Interface model demonstrated the potential utility of these manifold-based approaches, though with limited accuracy for trajectory prediction. These findings suggest that motor cortical activity evolves on curved, low-dimensional manifolds rather than linear subspaces.

[1]University of Pavia, FernUni Schweiz

[2]University of Bonn

[3]The Biorobotics Institute and Department of Excellence in Robotics and AI, Scuola Superiore Sant'Anna, Pisa, Italy

[4]International Centre for Translational Eye Research (ICTER), Institute of Physical Chemistry, Polish Academy of Science, Warsaw, Poland

1 Introduction

Neuroscience studies how the brain processes information to produce behavior, and studying how variables of the world are represented in neural activity is central to tackling this question. A topic of particular interest is the study of how the brain generates movement. This question is often explored by recording monkeys performing movements and observing their neural activity changes as a function of these movements.

In literature many studies highlight that the dynamics on the manifold offer a framework to interpret motor control that reflects the low-dimensional, nonlinear, and adaptive nature of neural motor planning.[4][6][3] The Neural manifold emerges from the connectivity of neurons that impose constraint on the neural activity that the network can generate and on what can be learned[13][10]. For extracting this manifold many linear[7] or non-linear[5] dimensionality reduction techniques were employed to detect patterns of co-activation of neurons (usually called neural "modes", for a review on dimensionality reduction see [4]).

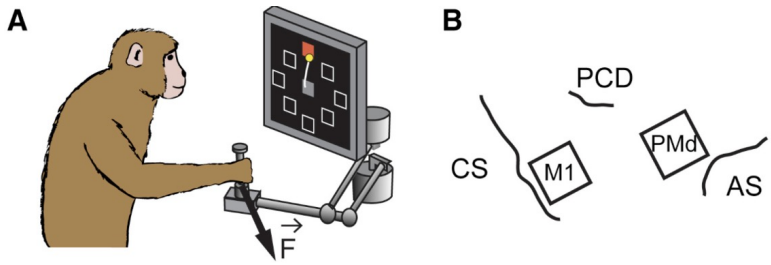


Figure 1: **A.** A monkey was trained to move a joystick with their hand to reach a target on the screen. **B.** In the meantime, populations of neurons were recorded in brain areas associated to movement.

In this study, we analyzed neural recordings from the primary motor cortex (M1) of a monkey performing a movement task(Figure 1). The task involved moving a cursor to one of eight target positions using a joystick. We focused on neural activity around movement onset ($t \in [-0.4, 0.8]s$), first examining the tuning of individual neurons to different reach directions.

We applied various dimensionality reduction techniques to uncover the underlying geometry of neural activity, including PCA and the recently developed MDDS. Our results revealed a curved, cone-like neural manifold with lower dimensionality than the number of recorded neurons. Leveraging these latent representations, we further explored their potential use for decoding or classify movements, testing their applicability for Brain-Computer Interface (BCI) applications.

Eventually we trained a Linear Dynamical Systems(LDS) to model the neural activity, evaluating its effectiveness as a surrogate for generating neural trajectories that are meaningful and retain information used for decoding the cursor's position.

Brain-Computer Interface

A Brain-Computer Interface (BCI) is an emerging technology that enables direct communication between the brain and external devices by translating neural data into machine-interpretable instructions. BCIs come in various forms, including invasive systems like Neuralink, which are surgically implanted on the brain's surface to allow direct control of machines through thoughts. However, non-invasive alternatives using EEG signals offer a less risky option for applications such as gaming and rehabil-

itation. In the context of this study, the BCI model should be able to control a joystick's direction based on visual targets displayed on a screen.

2 Methods

2.1 Neural Data

In this study we analyzed neural recordings during a movement task from a published work [11]. The subject – a monkey – is asked to move a cursor on the screen to reach one of eight target by moving a joystick. While the monkey performed the task, neural activity from 95 neurons was recorded from the primary motor cortex (M1).

2.2 Preprocessing

Neural activity was preprocessed to isolate task-related dynamics. Each trial generated a matrix $\mathbf{X} \in \mathbb{R}^{T \times N}$, where T denotes time points and N neurons, resulting in a 3D tensor $\mathcal{X} \in \mathbb{R}^{K \times T \times N}$ across K trials. Trials were grouped by target direction (8 conditions) and averaged per condition.

Spike trains and kinematic data were obtained from a dataset previously used in [11]. We focused on primary motor cortex (M1) activity. Spike trains were concatenated across channels and a zero-mean Gaussian filter ($\sigma = 4$ ms) was applied along the time axis to attenuate high-frequency noise.
Neural and kinematic signals were cropped to $[-0.4, 0.4]$ s relative to stimulus onset. Only trials belonging to the baseline (BL) epoch were retained. Neural data were baseline-corrected by subtracting the mean activity in the pre-stimulus window $[-0.4, 0)$ s and rescaled by the global standard deviation (with ϵ added for numerical stability).

2.3 Principal Component Analysis (PCA)

Following data preprocessing, we employed Principal Component Analysis (PCA) to characterize the neural population dynamics. PCA identifies orthogonal directions of maximal variance in high-dimensional data, enabling dimensionality reduction while preserving the most biologically relevant signals.

2.4 Uniform Manifold Approximation and Projection (UMAP)

Since the separation observed in PCA projections was not clearly distinguishable, we employed an alternative dimensionality reduction technique: Uniform Manifold Approximation and Projection (UMAP). UMAP is a powerful non-linear method particularly well-suited for visualizing high-dimensional data while preserving both local and global structures. Unlike PCA, which focuses on preserving linear relationships, UMAP maintains the underlying manifold geometry, making it especially effective for analyzing complex neural data.
The neural data originally resided in a high-dimensional space, with axes corresponding to neurons and time points. Reducing this dimensionality was essential to facilitate visualization and reveal potential patterns or clusters that are not apparent in the raw space. UMAP was selected for its ability to provide low-dimensional embeddings while preserving meaningful structure.

Once configured, UMAP was applied to the full neural dataset using the fit_transform method. This produced a two-dimensional embedding in which each point represents the high-dimensional neural activity at a specific time point, now projected into a 2D space.

To further examine condition-specific structure, we projected individual subsets, specifically conditions 0 and 1, using the pre-trained UMAP model. This ensured that the learned manifold structure was preserved across these subsets, enabling consistent comparisons.

Although Uniform Manifold Approximation and Projection (UMAP) provides a useful low-dimensional visualization of high-dimensional neural data, it fails to preserve the temporal continuity inherent in neural dynamics. Capturing these temporal aspects is critical for understanding the evolution of neural states during motor tasks. To overcome this limitation, we employed a more advanced approach—*Manifold Discovery via Dynamical Systems*, which integrates manifold learning with dynamical system modeling to capture both spatial and temporal patterns in the data.

2.5 UMAP Hyperparameter Search

To optimize the performance of UMAP for dimensionality reduction, a hyperparameter search was conducted using Optuna. This process is essential as UMAP's effectiveness significantly depends on several key parameters that influence the quality of the resulting embeddings.

The hyperparameters tuned include:

- `n_neighbors`: The number of neighbors considered in local manifold approximation, ranging from 5 to 50.
- `min_dist`: The minimum distance between points in the low-dimensional space, from 0 to 0.5.
- `n_components`: The dimensionality of the output space, explored between 2 and 10.

To ensure robust evaluation, a 5-fold cross-validation approach was employed. For each fold, UMAP was applied to transform the training data, followed by a validation step where a Linear Regression model predicted outcomes from the transformed features. This setup allowed for an unbiased assessment of the embedding quality across different splits.

The performance metric used for the optimization was the R2 score, which measures the proportion of variance explained by the model. The mean R2 score across all folds determined the best configuration, ensuring that the selected hyperparameters consistently performed well across different subsets of data.

2.6 Manifold Discovery via Dynamical Systems (MDDS)

To explicitly model temporal evolution, we applied Manifold Discovery via Dynamical Systems (MDDS). This approach merges manifold learning with dynamical system modeling, allowing us to recover both the low-dimensional manifold structure and the underlying temporal trajectories of neural activity. The central assumption is that neural responses evolve along smooth, low-dimensional manifolds governed by latent dynamics that can be described by differential equations. These parameters control various aspects of the model training, including the manifold and embedding dimensions, the penalty for regularization, the batch size, and the learning rate, among others. Specifically:

- Manifold dimension: 2 (to evaluate performance in a theoretically optimal low-dimensional space)
- Embedding dimension: 12 (to allow flexibility in capturing higher-order dynamics)

The model was trained using the `mdds.fit()` function, which optimizes the manifold representation by minimizing the reconstruction error and ensuring that the dynamics of the system are appropriately captured. The training process included regularization through the Frobenius penalty, which discourages overfitting by penalizing large deviations from the expected structure of the dynamics.
An important aspect of this analysis is to consider why a 2-dimensional manifold might be a theoretically sound choice for representing the data. From a theoretical standpoint, a 2D manifold is often sufficient for capturing the most significant features of high-dimensional data, particularly when the data lies on a lower-dimensional curved space. In the context of neural data, a manifold of higher dimensionality might introduce overfitting, potentially capturing noise rather than meaningful structure. Conversely, a 2-dimensional manifold may allow for the most relevant variability in the data to be captured, while excluding irrelevant fluctuations.

2.7 Classification of reached target

A Linear classification (Linear Discriminant Analysis - LDA) of the target angle was performed using data at the final time point, when the target is reached. The classifier was tested following a stratified 5-fold cross-validation scheme. The Matthews correlation coefficient[2]($\rho_M \in [-1, 1]$) was used as metric since it is more suitable for unbalanced datasets, as ours. Different neural data representations (raw, PCA, UMAP, MDDS) were used to evaluate which representation makes the information about the target angle linearly accessible.

2.8 Linear regression of cursor position

The dataset was composed also by the (x, y) position of the joystick used by the monkey to move the cursor on the screen (Figures 1 and 6a). We trained a linear regression model with least square to infer the original kinematic data for each time point (T samples, stored in a matrix $Y \in \mathbb{R}^{T\times 2}$) given the actual neural data representation (either raw, PCA, UMAP or MDDS).

2.9 Brain-Computer Interface Model

A brain computer interface model that would be applicable to this task is one where it can predict the next joystick position x_{t+1} given the previous position x_t. A minimal way in which this could be accomplished is with a linear model of the form

$$x_{t+1} = Ax_t \tag{1}$$

One way to train this model is to use the dynamics of the neural data $\{N_t\}$ as a proxy through the use of a decoder that can map neural data into position, as illustrated in figure 2. We have used a feed-forward model to accomplish this mapping.

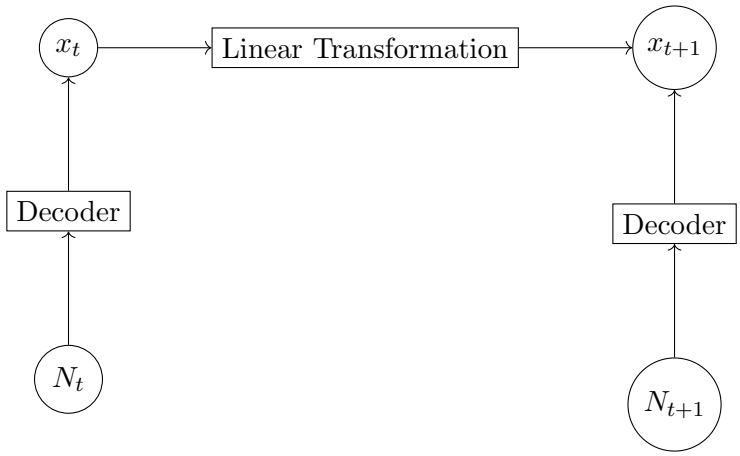


Figure 2: Schematic for the training of the brain-computer interface model using the neural data for proxy learning.

A minimal Brain-Computer Interface (BCI) model that can predict the next joystick position x_{t+1} based on the previous position x_t can be achieved through a linear transformation of the form

$$x_{t+1} = Ax_t \tag{2}$$

Where A represents the transformation matrix. To train this model, we leverage the dynamics of neural data $\{N_t\}$ as a proxy. This is achieved through the use of a decoder that maps neural activity to joystick position, as illustrated in Figure 2. Specifically, we employ a feed-forward neural network to establish this mapping. The transformation matrix A is subsequently learned via linear regression.

3 Results

3.1 Single unit's tuning curve analysis

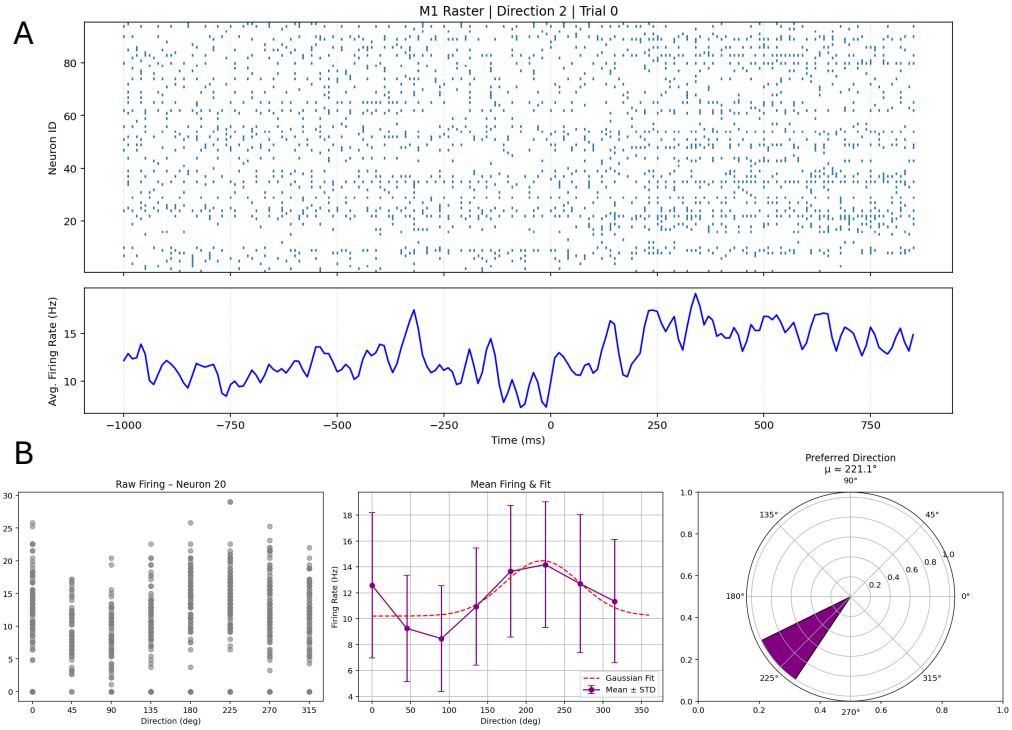


Figure 3: **Raster Plot and Population Firing Rate (Direction 2, Trial 0) and Direction Tuning for a Neuron. (A)** Top: Raster plot of 96 neurons. Bottom: Population average firing rate (bin-smoothed, Hz) across the trial. **(B)** Direction Tuning for Neuron 20. Raw firing rate across directions. Directional tuning curve with Gaussian fit. Polar plot of preferred direction.

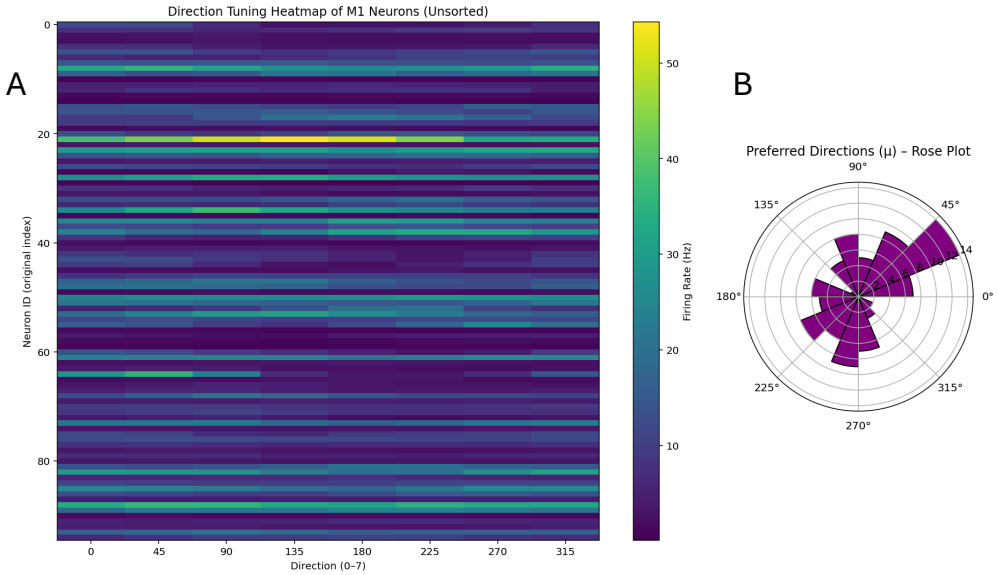


Figure 4: **Direction Tuning Heatmap and Distribution of Preferred Directions (Rose Plot).** Color scale = Firing rate (Hz). Each row = 1 neuron. Rose plot of tuning preference angles binned over 30° intervals.

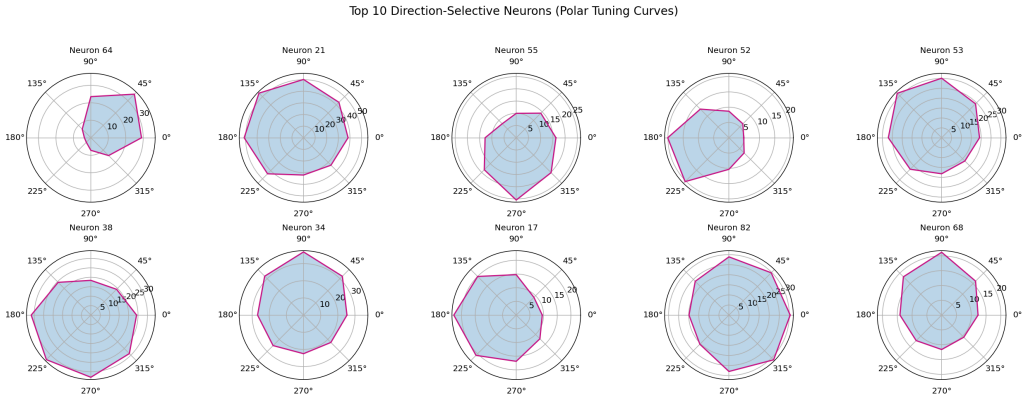


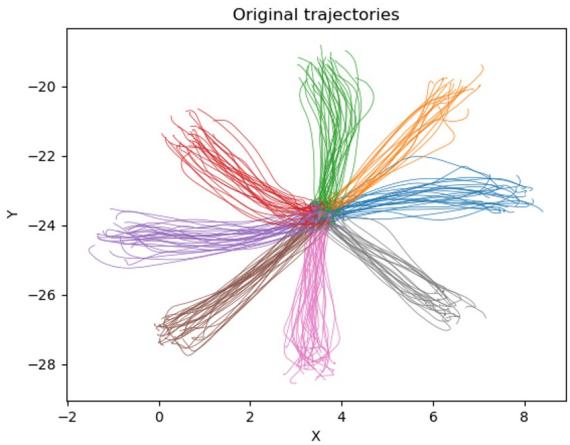
Figure 5: **Top 10 Direction-Selective Neurons (Polar Tuning Curves).** Each subplot: One neuron's polar tuning curve (max = peak response in Hz).

In Figure 3 shows the spiking activity of 96 M1 neurons during a single trial (Trial 0) in Direction 2 (target direction). The top panel presents a raster plot, where each row corresponds to a neuron and each dot marks a spike time. The bottom panel shows the average firing rate of the population over time, indicating modulations in activity around movement onset (time = 0 ms). Furthermore, Panel A in Figure 3 shows the raw firing rates of Neuron 20 across all directions and trials. Panel B displays the mean ± standard deviation of firing rates for each direction, with a Gaussian fit overlaid in red. Panel C is a polar representation indicating the preferred direction ($\mu \approx 221.1°$), suggesting a tuning toward that direction. In Figure 4, Firing rate responses of all M1 neurons (rows) are shown as a heatmap across 8 movement directions (columns). Each row corresponds to a neuron ID, and the color intensity represents the firing rate (Hz). Strongly tuned neurons display peak responses in specific

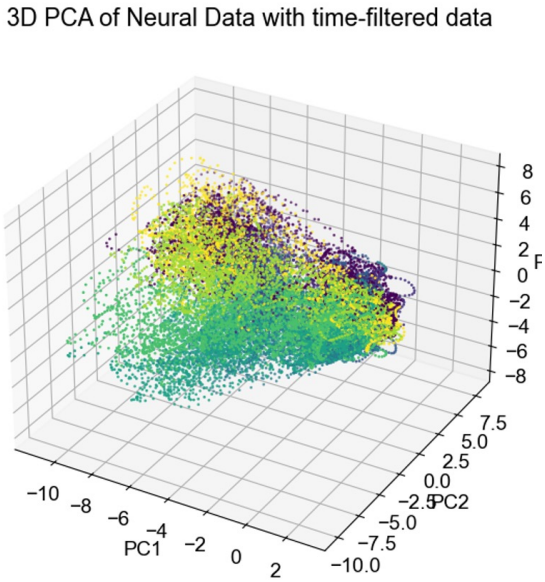
direction columns. Polar tuning curves for the top 10 direction-selective neurons. The blue-shaded area shows firing rate magnitude across 8 directions. These neurons exhibit clear peaks, each tuned to a specific preferred angle (Figure 5). A rose plot summarizing the distribution of preferred directions across the neural population. Each bar represents the number of neurons tuned to a specific angular bin, with a notable peak around 45° (Figure 4).

3.2 Dimensionality reductions techniques on Neural data

To assess whether distinct experimental conditions elicited differentiable motor patterns, we began with a qualitative examination of the kinematic data. We analyzed hand movement trajectories in the two-dimensional workspace, with each trajectory color-coded according to its corresponding experimental condition. These trajectories were extracted from a precisely defined time window aligned to the baseline (BL) epoch, ensuring a consistent comparison across trials while focusing on unperturbed movement execution. Figure 6a illustrates the resulting hand paths for all trials, with distinct colors representing the eight experimental conditions.



(a) Hand movement trajectories in the 2D workspace. Each trace represents the path followed during a single trial (n = 112 trials), colored by experimental condition (8 distinct targets). Trajectories were extracted during the baseline epoch (BL) and normalized to common start/end points.



(b) PCA applied to the full dataset, including all time points and conditions. The 3D plot shows the trajectories of neural activity over time, with each trajectory corresponding to a different condition.

Figure 6: Kinematic and neural trajectories during the center-out reach task.

We applied time-resolved PCA to investigate the temporal structure of neural dynamics throughout each trial. This analysis revealed condition-dependent trajectories in neural state space, supporting the idea of evolving, structured dynamics underlying the task (Figure 6b).

Although PCA revealed condition-specific dynamics, its linear nature limits its ability to uncover complex neural geometry. To overcome this, we applied UMAP, a nonlinear dimensionality reduction technique that preserves local neighborhood relationships and is more sensitive to intrinsic low-dimensional structure.

To explicitly capture the structure of neural trajectories over time, we employed the MDDS (Manifold-based Dynamical System Synthesis) method. MDDS infers low-dimensional latent dynamics from high-dimensional time series data, allowing us to model the temporal evolution of neural population activity. The results are summarized in Figure 7.

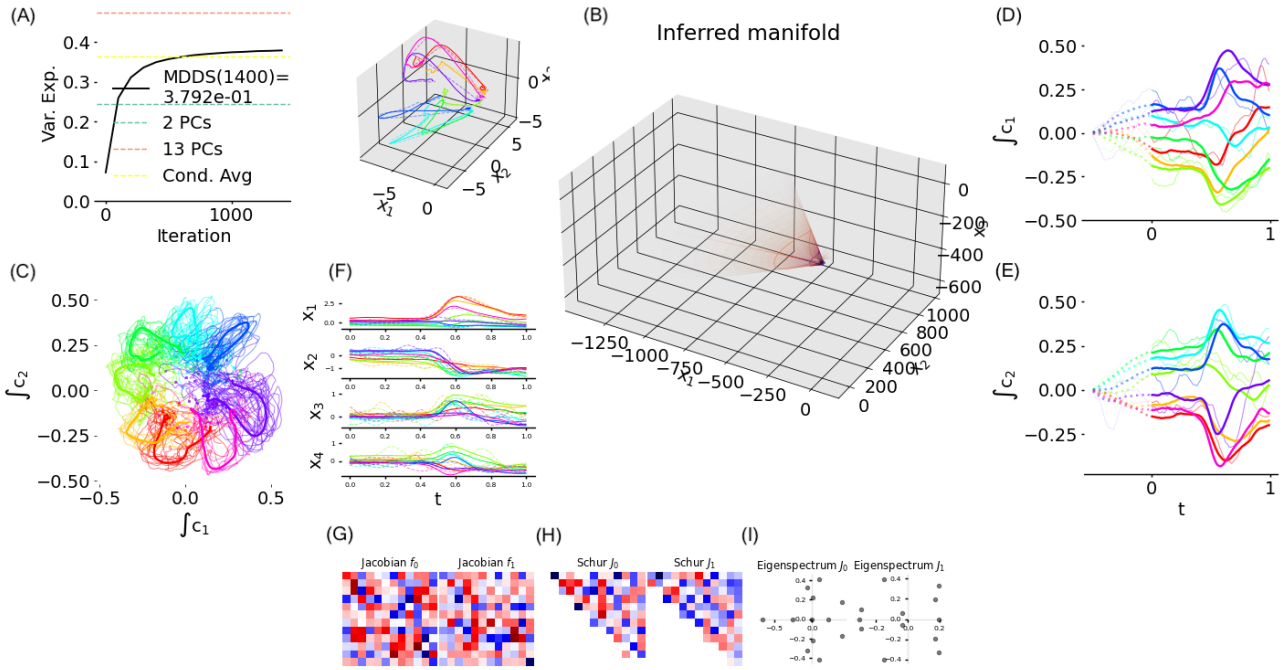


Figure 7: Dynamical manifold inference using MDDS. (A) Explained variance during training. (B) 3D trajectory plot and inferred latent trajectories. (C) Projections in the space of integrated control signals $\int c_1$ and $\int c_2$, colored by condition. (D,E) Time courses of integrated controls. (F) Reconstructed observed signals for selected dimensions. (G,H,I) Jacobians, Schur decompositions, and eigenvalue spectra of the learned dynamics.

In order, panel (A) shows the amount of explained variance across optimization iterations. While the first two principal components (PCs) account for only a limited portion of the variance, the 2D latent space inferred by MDDS explains approximately 38% of the total variance, outperforming linear PCA of the same dimensionality. This demonstrates the benefit of using nonlinear dynamic models for dimensionality reduction.

In Panel (B), the low-dimensional latent dynamics inferred by MDDS for each condition reveal structured and condition-specific trajectories in the learned latent space. The manifold exhibits a cone-like geometry, suggesting that the dynamics expand from a narrow initial region into condition-specific trajectories, following a smooth and continuous structure. This conical shape suggests that trajectories start from a similar region and then diverge according to well-structured dynamic rules, consistent with the hypothesis that the observed behavior has an intrinsic low dimensionality, well represented in 2D but embedded in a broader dynamic space.

Panel (C) displays the trajectories in the space of time-integrated control inputs $(\int c_1, \int c_2)$. The

clear separation between conditions indicates that the latent dynamics are not only structured but also discriminative. The fact that the trajectories are well-separated and structured suggests that the latent dynamics captured by MDDS are consistent with the experimental conditions, offering sharper separation compared to static methods like PCA or UMAP.
Panels (D) and (E) present the temporal evolution of the integrated controls $\int c_1(t)$ and $\int c_2(t)$, respectively. The distinct time courses further confirm that the inferred dynamics are condition-specific and evolve consistently across trials.
Panel (F) shows the reconstructed time series of selected observed variables, demonstrating that the learned latent dynamics successfully capture the main features of the original data.
Finally, Panels (G-I) provide a dynamical analysis of the system through the estimated Jacobians, Schur decompositions, and eigenvalue spectra of the learned dynamical system. The eigenvalue distributions suggest a mixture of stable and weakly unstable modes, consistent with a dynamical system that supports rich yet structured temporal evolution.

3.3 UMAP Hyperparameter Search

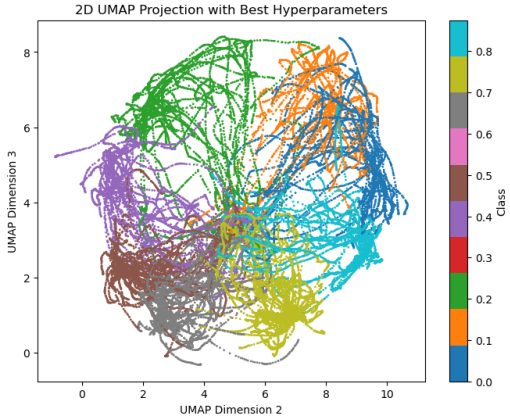


Figure 8: UMAP Embedding of the neural data after hyperparameter optimization with Optuna.

The optimal hyperparameters for UMAP embedding identified after 20 trials in Optuna are `n_neighbors`= 24, `min_dist`= 0.129, `n_components`=9. While Optuna's hyperparameter search does not guarantee finding globally optimal values, it successfully identified parameters that yield a meaningful representation of the data. The resulting UMAP embedding is depicted in Figure 8, demonstrating its ability to separate the 8 target angles despite having no information about them during the embedding process.

3.4 MDDS can denoise neural data for Classification tasks

We evaluated an LDA classifier's ability to decode the target angle using neural activity representations at the final time point of the monkey's reach. Successful decoding ($\rho_M > 0$) indicates that neural activity at this stage encodes the reached target angle. The LDA was trained separately on the representations generated using various dimensionality reduction techniques (Raw, PCA, UMAP, MDDS), as detailed in Methods. Comparing classification performances across these methods reveals which representation is most suitable for making the encoded target angle information linearly separable for the classification task.

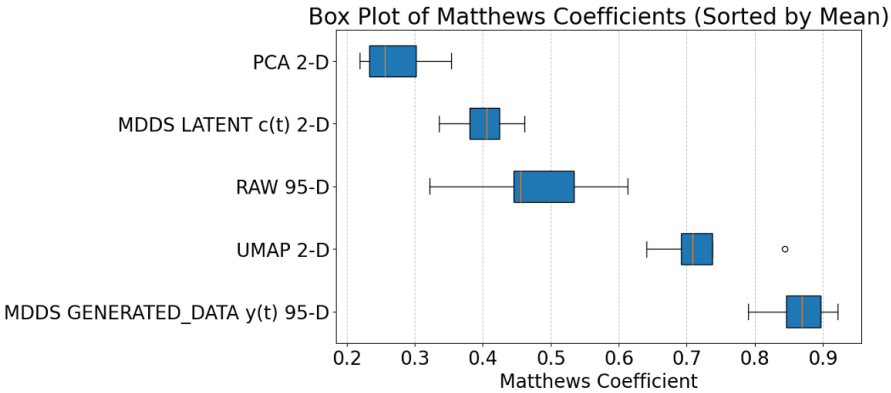


Figure 9: Box plot of Matthews Correlation Coefficients: LDA classification of the target angle using different neural data representations (RAW, PCA, UMAP, MDDS latent, and MDDS generated data) at the final time point. The results are sorted by the mean coefficient.

The classification results presented in Table 1 and Figure 9 shows that, for the same number of retained dimensions (95-D), the MDDS-generated representation consistently outperforms the raw neural data, as indicated by the higher Matthews correlation coefficient ($\rho_M \approx 0.87$ for MDDS against $\rho_M \approx 0.47$ for RAW data). This result suggests that the MDDS method effectively enhances the linear separability of the neural data, due to the latent-model representation.

The best low-dimensional (2-D) representation of neural data for linear classification of target angle using the last time point was achieved by an optimized UMAP ($\rho_M \approx 0.74$).

Using techniques that reveal non-linear manifolds (MDDS and UMAP) improves classification performance even using the same number of dimensions. This suggests that the manifold for discriminating motor commands used in this task, in motor cortex, is curved. The task-relevant neural activity crucial for decoding is likely evolving on a non-linear curved manifold.

Table 1: Classification results using different neural data representation. Rows are sorted first by the number of retained dimensions and then by classification performance.

Neural representation type	**Matthews Coefficient,** $\rho_M \in [-1, 1]$	**Std Matthews Coeffs**
MDDS Generated data $y(t)$ 95-D	**0.865108**	0.0452809
RAW Neural data 95-D	0.473754	0.0972355
UMAP 2-D	**0.740427**	0.0625496
MDDS Latent trajectories $\int c(\tau)d\tau$ 2-D	0.401399	0.0420250
PCA 2-D	0.272907	0.0493123

3.5 Regression of kinematic data using linear regression

Linear regression of primary motor cortex (M1) neuronal activity to kinematic parameters (i.e. position) is of interest for both Brain-Computer Interfaces (BCI) and general Neuroscience. It is important to study how different motor commands and movement parameters are represented in neuronal population activity[1][9]. For BCI application, the ability to decode kinematic intent from the neural

signal is crucial for enabling prosthetic control or cursor movement on a screen [8] [6] [12] [14]. Table 2 reports the results for linear regression of kinematic data

Table 2: R^2 Scores (Mean and Std) using different neural data dimensionality reduction techniques.

#	**Name**	$\mathbf{R^2}$ **Mean**	$\mathbf{R^2}$ **Std**
1	MDDS Generated data $y(t)$ 95-D	0.662667	0.00668309
2	RAW Neural data 95-D	0.565325	0.00737082
3	PCA 3-D	0.148393	0.00758670
4	MDDS Latent trajectories $\int c(\tau)d\tau$ 2-D	0.120509	0.00469726
5	UMAP 2	0.106768	0.00207547

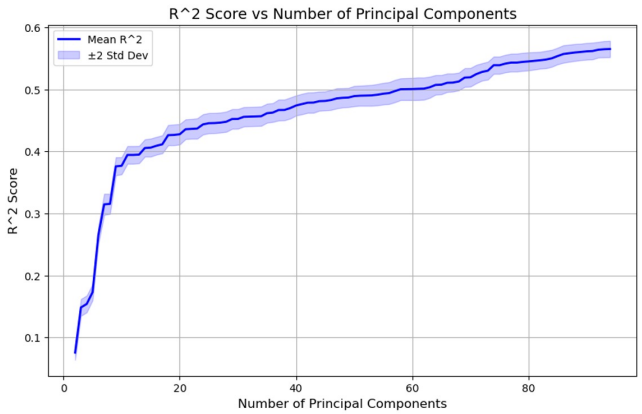


Figure 10: Performances on the regression of kinematic(R^2-score varying the number of principal components used in input.

In Figure 10 we observe how well the cursor (x,y) position can be approximated with linear regression as a function of the number of principal components (PCs) of the neural data. The R^2-score, the percentage of variance in kinematics explained by the regression, rises very steeply initially with increasing numbers of PCs added. But after some 20-30 principal components, the curve begins to flatten, approaching as expected the same performances of using the RAW neural data 95-D (see Table2). The presence of such a plateau means that adding more than a certain number of principal components does not significantly improve the linear decoding of kinematics variable. This confirms the findings in literature that neural activity in M1 relies on a manifold with a dimensionality lower than the number of recorded units [9] [6][1].

The best-performing regressor uses MDDS-generated 95-D data, outperforming raw neural data of the same dimensionality in linear regression of kinematic variables. This suggests MDDS effectively denoises neural signals, retaining only task-relevant information of the trajectory.

3.6 Brain-Computer Interface Model

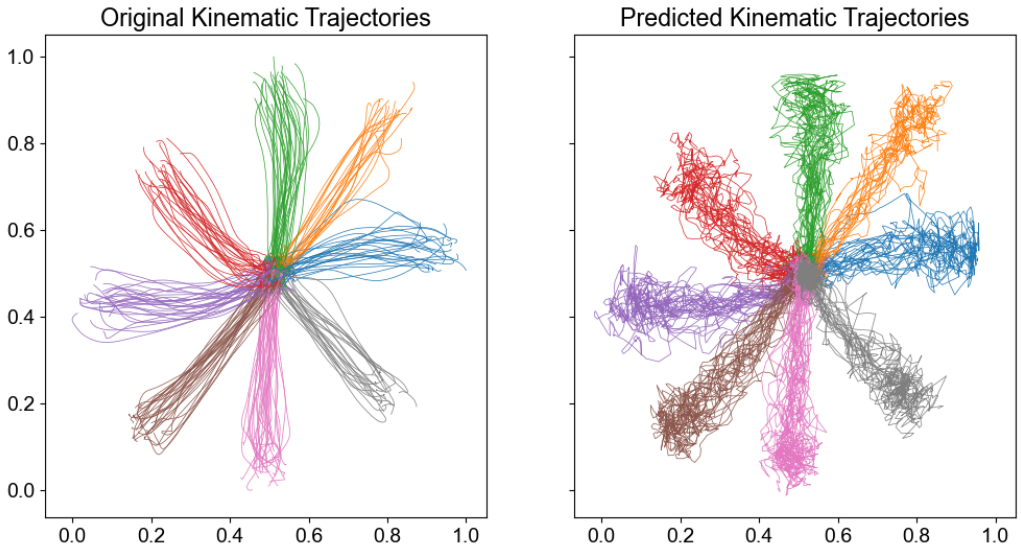


Figure 11: A comparison of original and predicted kinematic trajectories utilizing our BCI model.

The results of the BCI model described in Section 2.9 are presented in Figure 11. As illustrated in this figure, the predicted kinematic trajectories exhibit a chaotic nature and do not closely align with the original behavior of the data. Despite this discrepancy, the employment of a non-linear feed-forward decoder enables the mapping of these trajectories into a space that bears resemblance to the original one.

This approach highlights the challenges in accurately predicting complex behaviors using current BCI methodologies while also showcasing the potential of non-linear decoders in approximating such dynamics.

4 Conclusions

In this study, we demonstrate that motor cortex population activity during a center-out reach task evolves on a low-dimensional, curved manifold. Applying non-linear dimensionality reduction (MDDS) yielded a cone-like manifold that both denoised the neural signals and let the target direction information more linearly separable —outperforming PCA, UMAP, and raw data in LDA classification ($\rho_M \approx 0.87$(MDDS) vs. $\rho_M \approx 0.47$(Raw neural data) at 95-D). Similarly, MDDS-generated representations improved linear regression of joystick kinematics ($R^2 \approx 0.66$ (MDDS) vs. $R^2 \approx 0.57$ (Raw neural data)). Together, these results provide evidence that MDDS is a good dimensionality technique for retaining task-relevant information. Furthermore, neural dynamics relevant for both decoding discrete targets and continuous movement variables are constrained to a curved manifold rather than a flat subspace, see comparison between PCA and MDDS performances in Table 1, 2.

We further show that a simple Linear Dynamical System trained on MDDS-denoised trajectories can recapitulate the observed neural trajectories, suggesting a minimal computational model in which preparatory activity initializes a point on the manifold and subsequent autonomous dynamics drive movement generation. For real-time Brain-Computer Interface applications, this implies that lever-

aging manifold-based latent factors can enhance both classification- and regression-based decoders, improving robustness and accuracy under linear readout constraints.

In this study, we investigated whether neural population activity recorded during a motor task exhibits condition-specific low-dimensional dynamics that reflect the structure of behavior. Through a combination of dimensionality reduction techniques and dynamical systems modeling, we provide converging evidence that distinct experimental conditions evoke differentiable neural trajectories embedded within a structured low-dimensional manifold.

Initial qualitative analysis of kinematic data revealed that different task conditions reliably elicited distinct movement patterns. This behavioral differentiation was mirrored at the neural level: both PCA and UMAP uncovered condition-dependent structure in neural activity, with UMAP offering greater separability than linear methods. However, only the application of MDDS enabled explicit modeling of the underlying neural dynamics, revealing interpretable latent trajectories with distinct temporal signatures for each condition. Notably, the trajectories inferred by MDDS originated from a common initial state and diverged systematically, forming a cone-like geometry suggestive of structured control processes.

These findings support the view that motor cortical activity is governed by condition-specific low-dimensional dynamics, which can be uncovered and characterized using nonlinear dynamical systems approaches. The MDDS framework, in particular, offers a promising avenue for extracting interpretable and behaviorally relevant structure from neural time series.

References

[1] Mehdi Aghagolzadeh and Wilson Truccolo. Inference and decoding of motor cortex low-dimensional dynamics via latent state-space models. 24(2):272–282.

[2] Pierre Baldi, Søren Brunak, Yves Chauvin, Claus A. F. Andersen, and Henrik Nielsen. Assessing the accuracy of prediction algorithms for classification: an overview. *Bioinformatics*, 16(5):412–424, 05 2000.

[3] Mark M. Churchland, Byron M. Yu, John P. Cunningham, Leo P. Sugrue, Marlene R. Cohen, Greg S. Corrado, William T. Newsome, Andrew M. Clark, Paymon Hosseini, Benjamin B. Scott, David C. Bradley, Matthew A. Smith, Adam Kohn, J. Anthony Movshon, Katherine M. Armstrong, Tirin Moore, Steve W. Chang, Lawrence H. Snyder, Stephen G. Lisberger, Nicholas J. Priebe, Ian M. Finn, David Ferster, Stephen I. Ryu, Gopal Santhanam, Maneesh Sahani, and Krishna V. Shenoy. Stimulus onset quenches neural variability: a widespread cortical phenomenon. 13(3):369–378. Publisher: Nature Publishing Group.

[4] John P. Cunningham and Byron M. Yu. Dimensionality reduction for large-scale neural recordings. 17(11):1500–1509. Publisher: Nat Neurosci.

[5] Cátia Fortunato, Jorge Bennasar-Vázquez, Junchol Park, Joanna C. Chang, Lee E. Miller, Joshua T. Dudman, Matthew G. Perich, and Juan A. Gallego. Nonlinear manifolds underlie neural population activity during behaviour. page 2023.07.18.549575.

[6] Juan A. Gallego, Matthew G. Perich, Lee E. Miller, and Sara A. Solla. Neural manifolds for the control of movement. 94(5):978–984. Publisher: Cell Press.

[7] Juan A. Gallego, Matthew G. Perich, Stephanie N. Naufel, Christian Ethier, Sara A. Solla, and Lee E. Miller. Cortical population activity within a preserved neural manifold underlies multiple motor behaviors. 9(1):4233. Publisher: Nature Publishing Group.

[8] Leigh R. Hochberg, Daniel Bacher, Beata Jarosiewicz, Nicolas Y. Masse, John D. Simeral, Joern Vogel, Sami Haddadin, Jie Liu, Sydney S. Cash, Patrick van der Smagt, and John P. Donoghue. Reach and grasp by people with tetraplegia using a neurally controlled robotic arm. 485(7398):372–375, 2012.

[9] Svenja Melbaum, Eleonora Russo, David Eriksson, Artur Schneider, Daniel Durstewitz, Thomas Brox, and Ilka Diester. Conserved structures of neural activity in sensorimotor cortex of freely moving rats allow cross-subject decoding. 13(1):1–14. Publisher: Nature Publishing Group.

[10] Emily R. Oby, Alan D. Degenhart, Erinn M. Grigsby, Asma Motiwala, Nicole T. McClain, Patrick J. Marino, Byron M. Yu, and Aaron P. Batista. Dynamical constraints on neural population activity. 28(2):383–393. Publisher: Nature Publishing Group.

[11] Matthew G. Perich, Juan A. Gallego, and Lee E. Miller. A neural population mechanism for rapid learning. *Neuron*, 100(4):964–976, 2018.

[12] J. Andrew Pruszynski and Jörn Diedrichsen. Reading the mind to move the body decoding neural signals of intention and movement should guide the development of neural prosthetics. 348(6237):860–861. Publisher: American Association for the Advancement of Science.

[13] Patrick T. Sadtler, Kristin M. Quick, Matthew D. Golub, Steven M. Chase, Stephen I. Ryu, Elizabeth C. Tyler-Kabara, Byron M. Yu, and Aaron P. Batista. Neural constraints on learning. 512(7515):423–426. Publisher: Nature.

[14] Sergey D. Stavisky, Jonathan C. Kao, Paul Nuyujukian, Stephen I. Ryu, and Krishna V. Shenoy. A high performing brain-machine interface driven by low-frequency local field potentials alone and together with spikes. 12(3). Publisher: J Neural Eng.